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Article

Integrating Physics-Based Simulations with Data-Driven Deep Learning Represents a Robust Strategy for Developing Inhibitors Targeting the Main Protease

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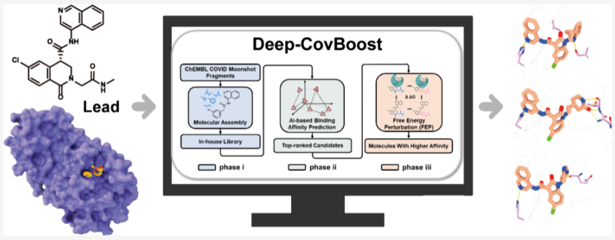
ABSTRACT: The coronavirus main protease, essential for viral replication, is a well-validated antiviral target. Here, we present Deep-CovBoost, a computational pipeline integrating deep learning with free energy perturbation (FEP) simulations to guide the structure-based optimization of inhibitors targeting the coronavirus main protease. Starting from a reported noncovalent inhibitor, the pipeline generated and prioritized analogs using predictive modeling, followed by rigorous validation through FEP and molecular dynamics simulations. This approach led to the identification of optimized compounds (e.g., I3C-1, I3C-2, I3C-35) that enhance binding affinity by engaging the underexploited S4 and S5 subpockets. These results highlight the potential of combining physics-based and AI-driven approaches to accelerate lead optimization and antiviral design.



1. INTRODUCTION

The COVID-19 pandemic was caused by the infection and spread of Severe Acute Respiratory Syndrome Coronavirus 2 (SARS-CoV-2), resulting in millions of deaths worldwide.[1,2] Since the onset of the 21st century, β-coronaviruses have been responsible for two other global pandemics and regional endemics, including Severe Acute Respiratory Syndrome (SARS) in 2003[3] and Middle East Respiratory Syndrome (MERS) in 2012.[4] Despite rapid progress in vaccine development and the approval of several directly acting oral antivirals for use against SARS-CoV-2 infection, such as ritonavir-boosted nirmatrelvir,[5] ensitrelvir,[6] and molnupiravir,[7] COVID-19 remains a threat, with the dominant Omicron variant continuously exhibiting immune evasion.[8,9] To proactively address potential future outbreaks of novel coronaviruses, there is still an urgent need to develop rapid-response technologies and potential coronavirus antivirals.

The coronavirus main protease serves as an attractive target for the development of coronavirus antivirals because of its essential role in viral replication and high conservation across various coronaviruses[10−15] (see Figure 1A). This protease cleaves viral polyproteins pp1a and pp1ab into functional protein subunits essential for viral replication.[16] As shown in Figure 1B, the catalytic His41-Cys145 dyad is located in the active site of the main protease and recognizes 11 distinct motifs for cleavage.[17] The inhibitors capable of binding to this active pocket can inhibit coronavirus replication.[11,18,19] Several main protease inhibitors have been identified. Nirmatrelvir, the protease inhibitor in Paxlovid, is a peptidomimetic covalent inhibitor.[20] Ensitrevir (Xocova) is a nonpeptidomimetic, noncovalent inhibitor.[21,22] Both have achieved clinical success. For early usage against viral infection or even as a prophylactic measure among vulnerable populations, an antiviral medication must be orally bioavailable and have an outstanding safety profile. Considering the inherent challenges of transforming peptide-based compounds into oral medications[23] and the potential downstream risks associated with covalent inhibition,[24] noncovalent and nonpeptidomimetic inhibitors represent promising options. Although the first-generation oral main protease inhibitor, nirmatrelvir, has demonstrated clinical efficacy, the necessity for coadministration of ritonavir—a combination marketed as Paxlovid can limit its utility in vulnerable populations due to potential drug−drug interactions.[25,26] To address this, a fully open-science, crowdsourced, and structure-enabled drug discovery campaign against the main protease of SARS-CoV-2, termed COVID Moonshot, was launched.[27] The COVID Moonshot Consortium provides a plethora of compounds along with their biochemical activity data and has established a comprehensive, open-access, and intellectual property-free knowledge repository that will facilitate future efforts in anticoronavirus drug discovery.

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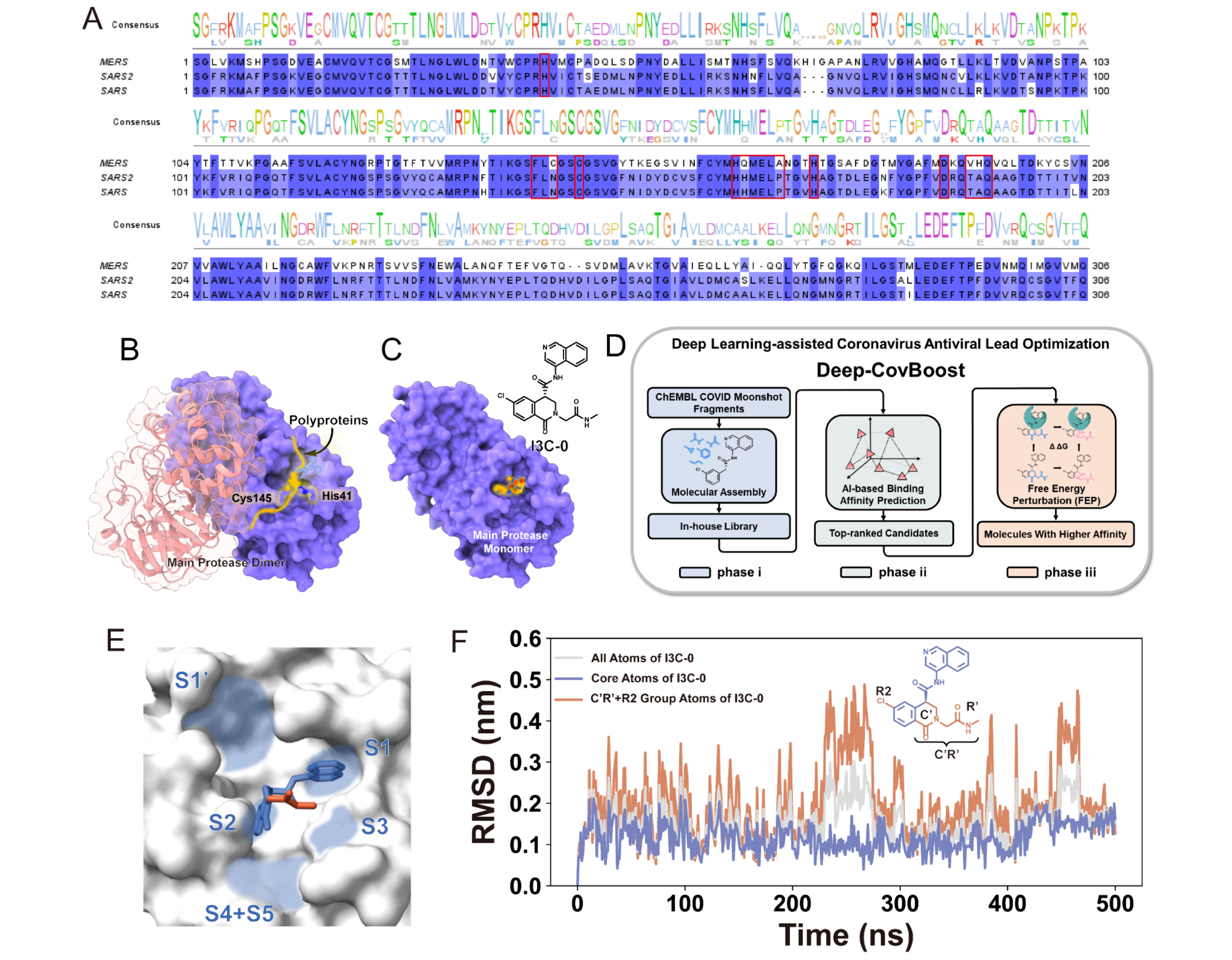

Figure 1. (A) Sequence alignment of the main proteases from SARS2 (PDBid: 7GLB), SARS (PDBid: 7DQZ), and MERS (PDBid: 9BOO), highlighting key residues in the active site (boxed in red). (B) Cartoon representation of the main protease dimer[35] (PDBid: 7P35) and its catalytic mechanism. The proximal monomer is depicted with secondary structure features and colored pink, while the distal monomer is illustrated with a purple surface. The active site at the distal monomer is highlighted by the presence of polyproteins in yellow with the catalytic dyad His41 and Cys145 labeled. (C) The crystal structure[27] of the complex formed between I3C-0 and the main protease (PDBid: 7GLB) is depicted. The main protease is represented with a purple surface, while the inhibitor is shown as a yellow stick. The molecular structure of I3C-0 is displayed in the upper right corner. (D) Illustrative workflow demonstrating the integration of physics-based FEP with data-driven deep learning in the Deep-CovBoost framework. (E) The binding mode of I3C-0 with the main protease and the distribution of subpockets in the active site of the main protease. (F) RMSD curves of I3C-0 in MD simulations. Three measurements of RMSD are displayed for parts of I3C-0: all atoms, core atoms (illustrated in blue), and C′R′+R2 group atoms (illustrated in red).

The crystal structure of the complex formed between the inhibitor with the highest potency from the COVID MoonShot (Moonshot ID: MAT-POS-a54ce14d-2, hereafter referred to as I3C-0, IC_{50} = 19.73 nM) and the main protease was determined through a collaboration between the Diamond Light Source XChem X-ray project and PostEra[27] (see Figure 1C). However, the binding affinity of nirmatrelvir (Moonshot ID: MAT-POS-7ed7af85-1, IC_{50} = 1.97 nM), which is FDA-approved, is 10-fold stronger than that of I3C-0. Similar to many other FDA-approved drugs, lead compounds typically undergo further optimization between their initial discovery and final approval before becoming available to the public. Improving the potency of leads is a critical step in this process. However, the vast chemical space, growing research and development (R&D) costs, and time-consuming development process are stubborn obstacles to boosting the output of novel drug-like molecules with desired affinities. To address this, recent studies have demonstrated that combining free energy perturbation (FEP) with machine learning strategies—such as active learning, reaction-based enumeration, and generative design—can accelerate lead optimization. For example, the combination of cloud-based FEP, reaction-based enumeration, and generative machine learning has facilitated the efficient exploration and prioritization of large, synthetically accessible chemical spaces.[28−30] Active learning strategies integrated with both absolute and relative FEP methods have demonstrated improved performance in virtual screening and lead optimization, particularly under data-scarce conditions.[31,32] In addition, the synergy between physics-based simulations and machine learning models has been shown to enhance the accuracy of binding affinity predictions.[33] Physics-based FEP and data-driven deep learning are two advantageous methods used for lead optimization. Deep learning techniques can rapidly explore the activity landscape of the chemical space

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surrounding lead compounds extending up to 10^{60}. And given a certain local chemical space, physics-based FEP can offer a high-precision scanning of free energy landscape. Inspired by these advances, we developed the Deep-CovBoost method (see Figure 1D), which seamlessly integrates physics-based FEP and data-driven deep learning for lead optimization. The Deep-CovBoost framework comprises three main phases. Briefly, (i) the binding stability analysis of I3C-0 reveals that the core scaffold atoms of I3C-0 demonstrate high binding stability. Therefore, these atoms should be preserved. In contrast, the RMSD analyses on C′R′ and R2 atoms of I3C-0 show that these groups are unstable in the binding pocket. This indicates that these atoms could be optimized to achieve a higher binging affinity (see Figure 1E,F). Molecular fragments obtained from the ChEMBL database[34] and the COVID Moonshot were utilized to replace unstable parts of I3C-0, thereby yielding in-house molecular libraries. (ii) Subsequently, a deep learning model, trained on the biological activity data of main protease inhibitors, was employed to evaluate the relative affinity between compounds in the in-house library and I3C-0. (iii) Finally, top-ranked compounds were subjected to FEP calculations and mechanistic analysis to identify drug candidates with anticoronavirus potential. This integrated strategy offers several advantages: (1) Deepening understanding: By developing a deep learning model, we achieve a deeper understanding of the intricate relationships between the structure and activity of coronavirus main protease inhibitors; (2) Efficient sampling: Through a combination of molecular generation and deep learning-based biochemical activity prediction, our approach enables rapid and extensive exploration of the activity landscape in chemical spaces; (3) Interpretable mechanism: Utilizing a combination of FEP calculations and extended Molecular Dynamics (MD) simulation, we gain valuable insights into the recognition mechanisms and molecular interactions that improve the efficacy of newly designed inhibitors targeting the main protease.

2. METHODS

2.1. Deep Learning. The SMILES of the main protease inhibitors were standardized using the *standardize_smiles* function from the MolVS software package.[36] Next, the atomic information and chemical bond information on the compounds were extracted using RDKit.[37] Various pieces of information were then input into neural networks for model training and prediction. All deep learning neural networks were constructed using PyTorch.[38]

Two performance indicators, viz., Mean Squared Error (MSE), and Pearson correlation coefficient (PCC) were used to evaluate the model for predicting.

$$\mathrm{MSE} = \frac{1}{N}\sum_{i=1}^{N}(x_i - y_i)^2 \tag{1}$$

$$\text{Pearson correlation coefficient (PCC)} = \frac{N\sum x_i y_i - \sum x_i \sum y_i}{\sqrt{N\sum x_i^2 - (\sum x_i)^2}\sqrt{N\sum y_i^2 - \left(\sum y_i\right)^2}} \tag{2}$$

where N represents the number of all samples, and x_i and y_i represent the labels and predicted values of samples, respectively.

2.2. Molecular Dynamics. Each system was solvated in a TIP3P[39] water box and then charged, neutralized, and ionized with 150 mM concentration using Na^+ and Cl^-. The resulting simulation systems contained roughly 48913 atoms. CHARMM36 force field[40] was used for the main protease protein, water and ions. CGenFF force field[41] was generated with the online server for I3C-0 and its analogues. NAMD3.0[42,43] was used to run all simulations. Particle mesh Ewald (PME)[44] was used for electrostatic interaction calculations with a grid size of 1 Å. A switching function was used for van der Waals (VDW) interaction calculations, where the switching distance was 10 Å, the cutoff distance was 12 Å, and the VDW pair list distance was 13.5 Å. A Langevin thermostat maintained the temperature at 310 K. Before production runs, 20000 steps of energy minimization and 10 ns of equilibration using 1 fs time step were performed. Production MD simulations were conducted for 500 ns using a 2 fs time step in triplicate. Snapshots of the trajectories were saved every 500 ps for data collection.

2.3. Free Energy Perturbation. The binding affinity changes, due to the substitution of fragments, between I3C-0 and the main protease were calculated using the FEP method. The dominant structures of the production MD simulations were selected for FEP computation. FEP protocol was adopted from a previous study.[45] Specifically, the complex structure of I3C-0 with the main protease was obtained from the RCSB, while the structures for I3C-1 through I3C-26, which were used for FEP calculations, were generated by editing the 3D structure of I3C-0 using PyMol. The free energy changes in both the bound state (I3C-0 and main protease complex) ΔG_{bound} and the free state ΔG_{free} were estimated using NAMD3.0. Thus, the binding free energy changes caused by the fragment substitution was estimated as $\Delta\Delta G_{\text{calc}} = \Delta G_{\text{bound}} - \Delta G_{\text{free}}$. CHARMM36 force field was used for the main protease protein, water, and ions, matching the MD work. For each fragment replacement, a dual-topology file was prepared based on the CGenFF force field, and 29 λ windows (0.00, 0.00001, 0.0001, 0.001, 0.01, 0.05, 0.1, 0.15, 0.2, 0.25, 0.3, 0.35, 0.4, 0.45, 0.5, 0.55, 0.6, 0.65, 0.7, 0.75, 0.8, 0.85, 0.9, 0.95, 0.99, 0.999, 0.9999, 0.99999, 1.00) with 2 ns/window were used. The van der Waals and electrostatic interactions were transferred simultaneously during the simulations. For each FEP calculation, three independent replicas starting from different conditions were performed for sufficient sampling, and at least 348 ns (2 ns × 29 windows × 3 runs × 2 states) simulation time was generated, which results in reasonable convergence in the free energy calculations.

2.4. Molecular Generator. All compounds in ChEMBL[34] were fragmented based on their synthetic routes using BRICS[46] in RDKit. A mapping function was established between the number of heavy atoms and the set of small molecular fragments according to the fragment information. A Python program was created to substitute specific fragments of compounds and generate an in-house small molecule library.

3. RESULTS AND DISCUSSION

3.1. Determining the Modification Hotspots in I3C-0 through MD Simulations. The mechanism of recognition of I3C-0 by the main protease plays a pivotal role in further lead optimization. As illustrated in Figure 1E, the active site of the main protease comprises six subpockets, denoted S1 to S5 and S1′.[18] The crystal structure of the complex reveals that I3C-0 occupies the S1, S2, and S3 subpockets, thereby inhibiting the

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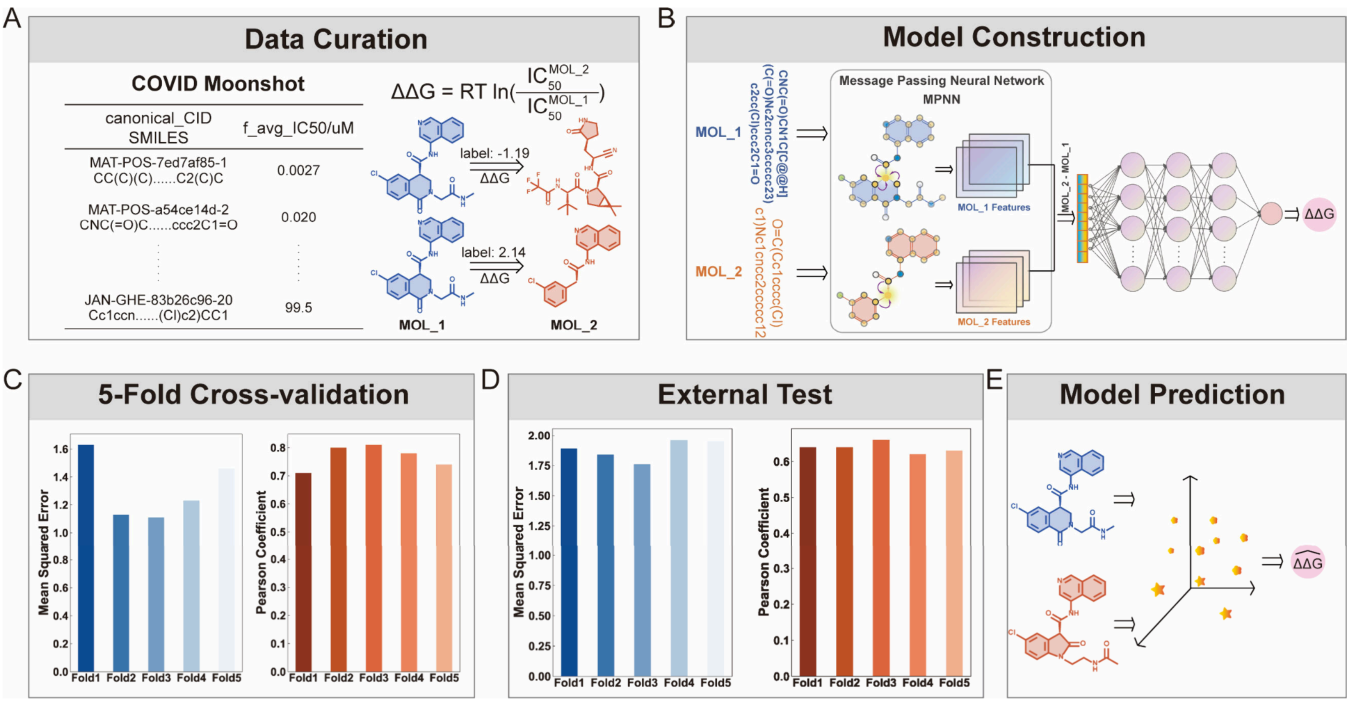

Figure 2. Workflow of constructing a model to assess the activity of small molecules targeting the main protease. It consists of five key steps: data curation (A), model construction (B), 5-fold cross-validation (C), external testing (D), and model prediction (E).

activity of the main protease. According to the findings of Zhang et al., the stability of ligand moieties bound to proteins affects their contribution to overall binding affinity.[45] Here, we employed RMSD calculations to qualitatively estimate the contributions of ligand moieties. As shown in Figure 1F, the blue-colored core atoms of I3C-0 exhibited higher stability upon binding to the protein in comparison with the red-colored C′R′+R2 group. These simulation findings are consistent with experimental observations, where the quinoline ring in I3C-0 forms hydrogen bonds with the ε-nitrogen on His163 in the S1 subpocket, and the carbonyl oxygen interacts with the Glu166 backbone nitrogen in the S3 subpocket[47] (see Table S1). These interactions stabilize the binding of the core structure of I3C-0 to the main protease. However, the red C′R′ group does not contribute much to binding to the protein. Through molecular modeling, two subpockets, S4 and S5, near the C′R′ group can accommodate the interactions. Therefore, a rational optimization strategy is to occupy the S4 and S5 subpockets with modified C′R′ group, thereby further enhancing the enzyme inhibitory activity of new developed analogs. Combining this strategy and the Deep-CovBoost framework, a molecular substitution program was employed to substitute the C′R′ group and R2 group, resulting in the creation of densely populated in-house libraries. Compounds with affinities higher than I3C-0 may exist within these molecular libraries. However, effectively and accurately picking them poses a significant challenge.

3.2. Development of a Deep Learning Model for Assessing Small Molecule Activity Targeting the Main Protease. A deep learning model that effectively integrates the prior knowledge of known main protease inhibitors' structure–activity relationships can quickly assess the activities of numerous known and newly designed molecules. The construction and evaluation of the model are illustrated in Figure 2. It consists of five key steps: data curation, model construction, 5-fold cross-validation, external test, and model prediction.

Data curation: The data set used for the deep learning model in this study was composed of all analogs sharing the same backbone as I3C-0 from the COVID Moonshot. This data set consists of 707 compounds, with inhibitory activities against the main protease ranging from 19.73 nM to 99.5 μM. To facilitate model validation and testing, the entire data set was split into training/validation and external testing sets in a 9:1 ratio. Then, all main protease inhibitors in the data set were paired, and their activity data were used in the form of binding free energy changes. Each entry in the data set contained a canonical SMILES pair and its corresponding binding free energy change. The training/validation set was divided into 5-fold cross-validation data, while the testing set served as external testing data. The data processing workflow is illustrated in Figure 2A.

Model construction: The deep learning model was developed to learn the patterns of structural differences among main protease inhibitors that led to differences in activity using molecular pairs as inputs. Small molecules were represented as molecular graphs,[48,49] where nodes represented atoms and edges represented chemical bonds. To enrich the representation of chemical features, molecular properties, such as atomic elements, atomic connectivity, formal charges, chirality, and aromaticity, were also included. Additionally, features of chemical bonds were used to enrich the edge information. However, these molecular graphs remained static representations, merely reflecting the topological structure of small molecules. To capture dynamic interactions among nodes, message passing neural networks (MPNNs)[50,51] were employed. This approach allowed for a different molecular graph generation between two structurally similar molecules. For example, two structurally similar molecules, MOL_1 and MOL_2, were compared in Figure 2B. The highlighted atom in MOL_1 showed significant differences in receiving

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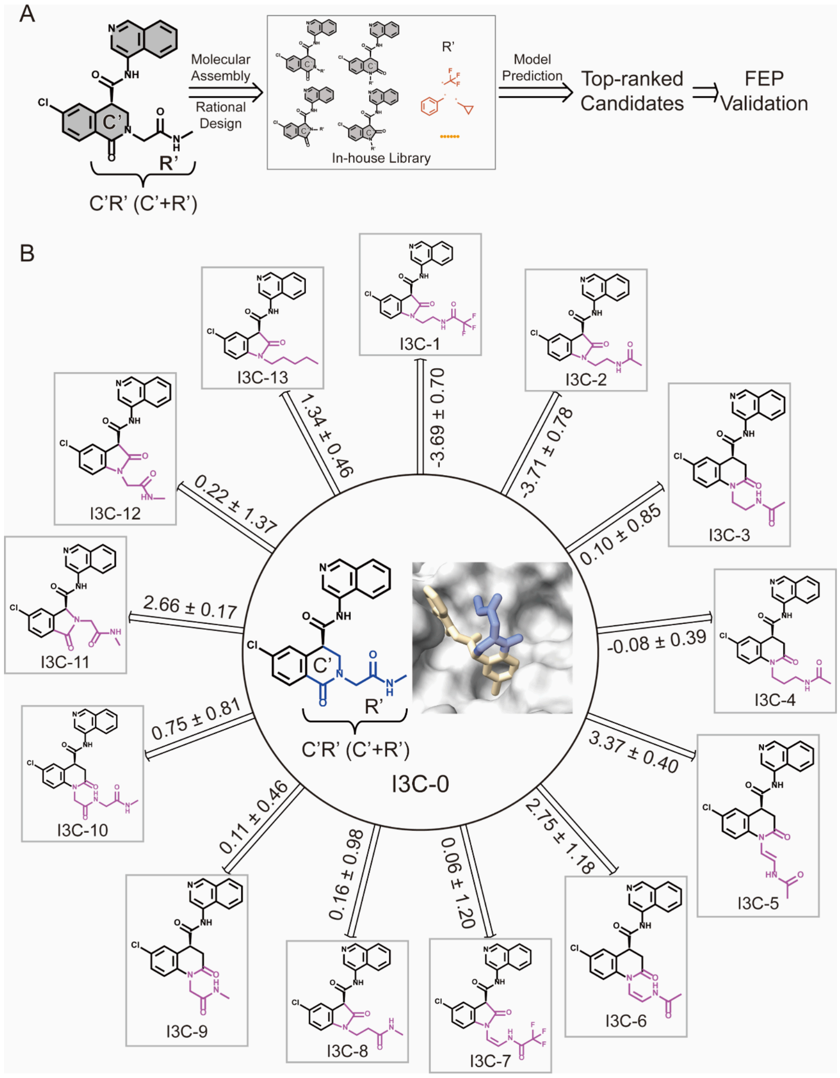

Figure 3. (A) Flowchart illustrates the integration of deep learning and FEP for lead optimization. (B) The FEP results for the optimization of the C′R′ group. The newly generated analogs denote I3C-1 to I3C-13.

information from its surrounding atoms compared to the corresponding atom in MOL_2. MPNNs extracted feature vector "MOL_1 feature" and "MOL_2 feature" for each molecule. These dynamic graphs sensitively represent structural differences. The differences in the feature vectors between two small molecules indicate their structural differences. A multilayer perceptron then established a connection between the molecular graph differences and the binding free energy change labels.

5-Fold cross-validation was employed for model training and hyperparameter tuning. The mean square error (MSE) and Pearson correlation coefficient between samples' labels and the model's predictions were utilized to assess the predictive performance of the model. As shown in Figure 2C and Table S2, the MSE of 5-fold cross-validation ranged from 1.11 to 1.63, and the Pearson correlation coefficient ranged from 0.71 to 0.81, indicating that structural differences in molecular graphs can effectively characterize changes in activity. To further assess the generalization performance of the model, MSE and Pearson correlation coefficients were also employed to evaluate the model's predictive performance on the independent external test set. As shown in Figure 2D and Table S3, the MSE ranged from 1.76 to 1.96, and the Pearson correlation coefficient ranged from 0.62 to 0.66. These results indicate that the model effectively learns the patterns of structural changes in main protease inhibitors that lead to alterations in binding free energy, demonstrating good extrapolation performance on a completely new data set. As shown in Figure 2E, the final model maps the functional relationship between structural changes and binding free energy changes for all known protease inhibitors. When predicting a novel compound, it is paired with I3C-0 and then input into the prediction model. The output value represents the predicted binding free energy change relative to I3C-0.

3.3. Validation of FEP Capabilities for the Complex between the Main Protease and I3C-0. The deep learning model can quickly perform a preliminary assessment of the activity of a large number of compounds, but its accuracy is limited by the richness of the data. The physics-based FEP is needed to further evaluate the affinities of top-ranked candidate compounds. FEP, one of the most rigorous and reliable methods for estimating binding affinity changes, has been successfully applied in numerous investigations involving protein−protein, protein−ligand, and protein−DNA interactions.[45,52−54] Here, to further validate the performance of the FEP method in the main protease and the I3C-0 binding complex, the binding free energy changes caused by fragment substitution in I3C-0 were investigated. In previous work, Boby et al. have explored the optimization pathway of I3C-0.[27] Four representative compounds from the reported optimization pathway were selected to test the FEP performance in this system (see Figure S1A). The inhibition efficacy (IC_{50}) of I3C-0 bound to the main protease and its analogs were experimentally determined in previous studies.[27,47] Then, the experimental relative binding free energy change caused by structural alteration can be derived as $\Delta\Delta G_{exp} = -k_B T \ln (IC_{50}^{wt}/IC_{50}^{mut})$, where IC_{50}^{wt} and IC_{50}^{mut} are inhibition efficacy of wide type (WT) and its analog, respectively, and k_B is the Boltzmann constant and T = 310.35 K. Thus, the four compounds in the lead optimization pathway served as an illustrative benchmark for testing the capability of *in silico* FEP method. There is a strong correlation between experimental measurements and FEP calculations for binding free energy changes due to fragment substitution (see Figure S1B,). Overall, our rigorous FEP calculations achieve high accuracy and robust performance in the main protease and I3C-0 binding system, comparable to experimental measurements, making its application in subsequent lead optimization credible.

3.4. Optimization of the C′R′ Group in I3C-0. Searching for small molecular analogs is important for identifying valuable candidates for developing inhibitors against coronavirus. Here, we try to develop I3C-0 analogs with comparable or even enhanced binding affinity using Deep-CovBoost. MD simulations reveal that optimizing I3C-0 to enable its C′R′ group to occupy the S4 and S5 subpockets is a primary strategy. However, the orientation of the I3C-0 R′ group is influenced by the connected six-membered lactam ring (C′ group). Thus, the size of the lactam ring and the position of the R′ group on the C′ group should also be taken into account. Four variants of the C′ scaffolds were used as starting structures to generate an analog library (see Figure S2). Then, as shown in Figure 3A, an in-house small molecule library containing 120000 analogs was generated by substituting the R′ group of the four scaffolds. Subsequently, these analogs were collectively input into the deep learning model to predict their relative activities compared to I3C-0. Following this, the top-ranked analogs (Score < −0.5) were qualified for FEP calculations to validate and rapidly identify inhibitors (see Figure S3 and Table S4). In the first round of Deep-CovBoost application, a total of 13 analogs met qualification criteria and were denoted as I3C-1 to I3C-13, as shown in Figure 3B. Notably, I3C-1 and I3C-2 exhibited significantly higher affinities compared to I3C-0. Compared with the structure of I3C-0, both I3C-1 and I3C-2 feature a pentacyclic ring in their C′ groups.

To gain deeper insights into the molecular mechanisms driving changes in binding affinity toward I3C-0 analogs resulting from structural modifications, MD simulations were performed for I3C-1, I3C-2, and I3C-0 in complex with the main protease. As shown in Figure 4, when the C′ group of I3C-0 was modified to a 5-membered ring, the extension direction of the R′ group of analogs tended to extend toward the S4 and S5 subpockets, allowing C′R′ group to occupy S4 and S5 subpockets, thereby enhancing the binding (see Figure S4). Importantly, the structural modification of the C′R′ group in I3C-1 and I3C-2 aligns with similarity with that of nirmatrelvir, the main component of the FDA-approved COVID-19 drug Paxlovid. This resemblance is not only

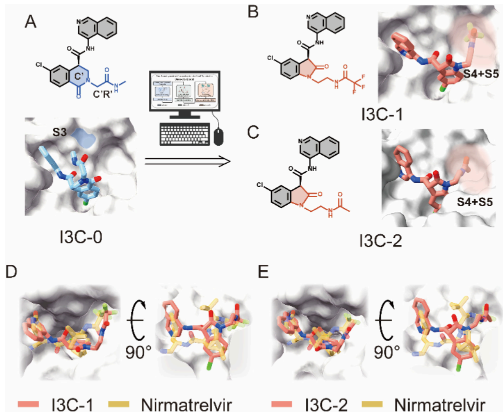

Figure 4. (A) C′R′ group of I3C-0 was optimized using the Deep-CovBoost framework. I3C-1 (B) and I3C-2 (C) with significantly enhanced affinity were successfully discovered. Comparison of the binding modes of I3C-1 (D), I3C-2 (E), and nirmatrelvir with the main protease.

evident in chemical structure (see Figure S5) but also reflected in their binding behavior toward the S4 and S5 subpockets. As shown in Figure 4, the binding modes of I3C-1 and I3C-2 with the main protease are similar to that of nirmatrelvir in S4 and S5 subpockets, indicating a resemblance in their binding modes. Further MD simulations revealed that nirmatrelvir, I3C-1, and I3C-2 all form stable interactions with key residues Leu167, Pro168, Thr190, Ala191, and Gln192 in the S4 and S5 subpockets of the main protease (see Figure S6). The contact frequencies of all three compounds with these critical residues were significantly higher than those of I3C-0. In summary, I3C-1 and I3C-2 with significantly enhanced affinity were successfully discovered through the Deep-CovBoost framework in this optimization process. The five-membered ring of C′ group of these compounds enables C′R′ to better occupy the S4 and S5 subpockets compared to I3C-0, thereby enhancing their binding affinities, laying the groundwork for further structural optimization.

3.5. Optimization of the R2 Group in I3C-0. Additionally, it is observed that most I3C-0 analogs reported by the COVID Moonshot consistently contain a chlorine atom. The Deep-CovBoost was then performed to validate the importance of the chlorine atom (see Figure S7). The study reveals that the binding affinity remains stable only when the chlorine atom is replaced with a bromine atom, whereas substitution with a fluorine atom or other fragments significantly decreases the binding affinity (see Figure S7, Figure S8, and Table S5). This finding suggests that the size and nature of halogen elements may play an important role in the S2 subpocket.

3.6. Optimization of I3C-2 to Fit the S4 and S5 Subpockets. Following the initial round of lead optimization, it is observed that converting the C′ group of I3C-0 from a 6-membered ring to a 5-membered ring enables its C′R′ group moiety to better extend into the S4 and S5 subpockets. Therefore, I3C-1 and I3C-2 were used as leads for a second round of optimization of the R5 group in the S4 and S5 subpockets. Through the Deep-CovBoost workflow (see Figure S9 and Table S6), a new set of candidates with binding affinities comparable to or even stronger than those of I3C-1/I3C-2 were indeed obtained. Compounds such as I3C-34 ($\Delta\Delta G = -1.36 \pm 0.32$ kcal/mol), I3C-35 ($\Delta\Delta G = -1.01 \pm$

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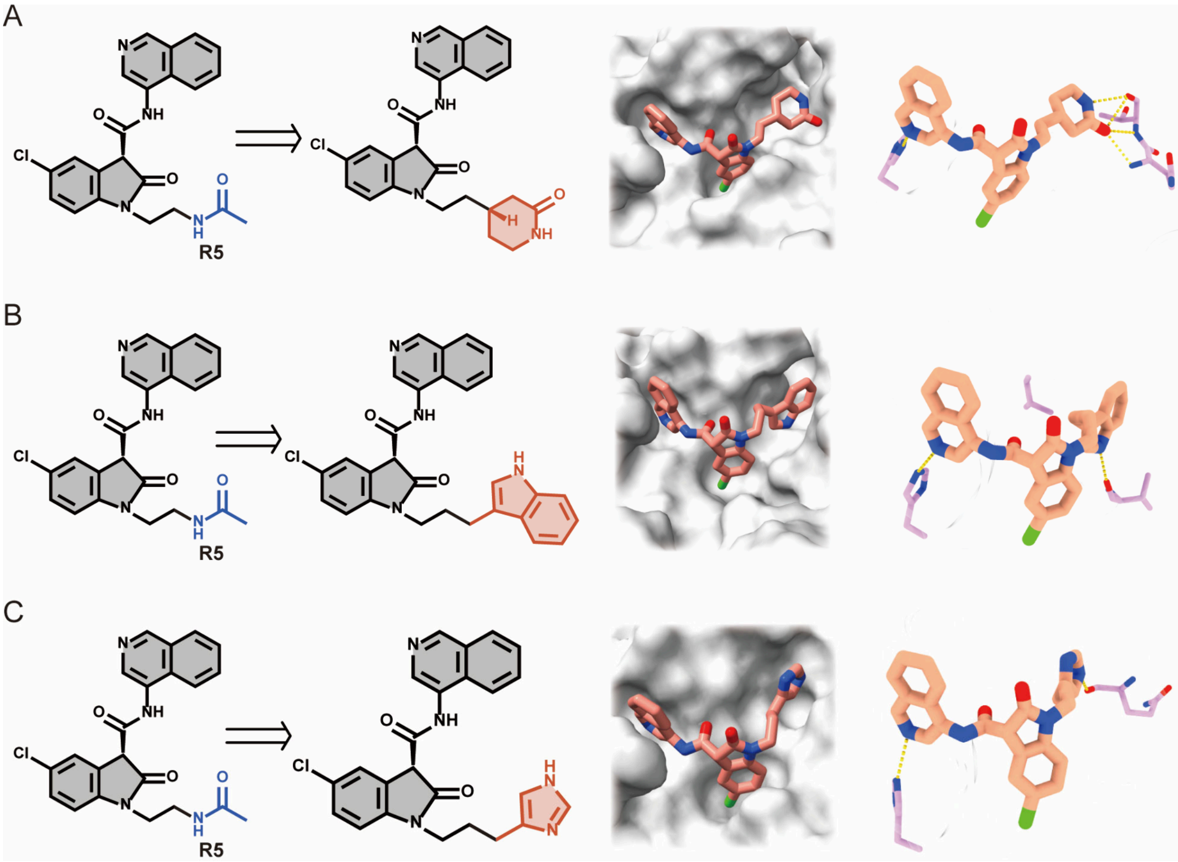

Figure 5. Molecular dynamics results and interaction analysis between I3C-34 (A), I3C-35 (B), and I3C-36 (C) and the main protease.

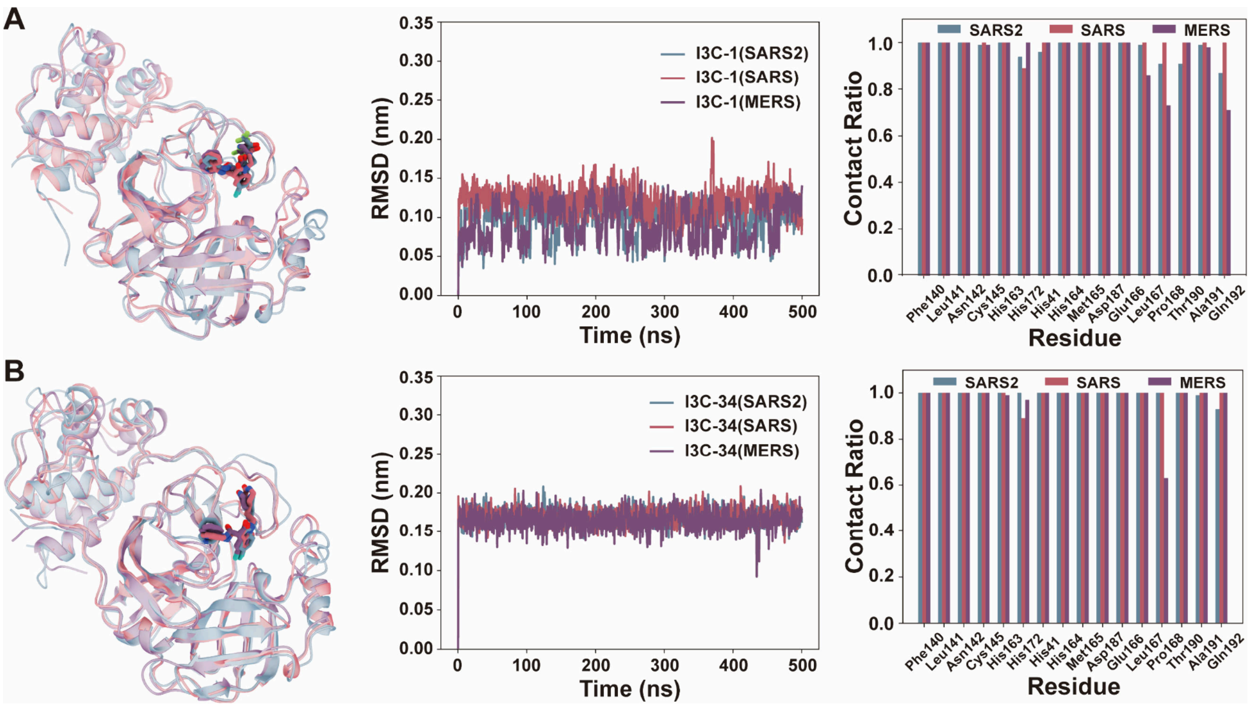

Figure 6. RMSD curves from MD simulations of I3C-1 (A) and I3C-34 (B) with the main proteases of SARS2, SARS, and MERS, along with the contact frequencies with key active site residues.

0.49 kcal/mol), and I3C-36 ($\Delta\Delta G = -0.64 \pm 0.12$ kcal/mol) demonstrate better fitting into the S4 and S5 subpockets compared to I3C-1/I3C-2 (see Figure S10).

To further explore the accommodation properties of the S4 and S5 subpockets for small molecular fragments, MD simulations were conducted for I3C-34, I3C-35, and I3C-36

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in complex with the main protease. As shown in Figure 5, these three compounds indeed exhibit better topological fitting within the S4 and S5 subpockets of the active site compared to I3C-1 and I3C-2. The lactam in the R5 group of I3C-34 forms a hydrogen bond network with the backbone of the S4 and S5 subpockets. Similarly, the indole and imidazole groups in the R5 group of I3C-35 and I3C-36 also form hydrogen bonds with the S4 and S5 subpockets. Moreover, a comparison with recently reported inhibitors[55] shows that I3C-35 shares similar fragment topology with experimentally validated compounds AC1115 and VK-13 (Figure S11), further supporting the reliability of both the compound design and the Deep-CovBoost framework.

3.7. Interaction Stability of I3C-1 and I3C-34 Binding to the Main Proteases of SARS2, SARS, and MERS. To evaluate the broad-spectrum inhibitory potential of candidates identified via the Deep-CovBoost framework, we performed MD simulations of I3C-1 and I3C-34 with the main proteases of SARS2, SARS, and MERS. These simulations demonstrated that both compounds exhibited stable binding in the active sites throughout the duration, with RMSD values in the range of approximately 0.5 to 1.5 Å for I3C-1 (see Figure 6A) and 1.5 to 2.0 Å for I3C-34 (see Figure 6B) across the three coronaviruses, indicating minimal structural fluctuations. Significantly, both I3C-1 and I3C-34 maintained strong affinity for the active sites of multiple coronaviruses, indicating their potential as versatile inhibitors. Analysis of interaction frequencies revealed that the optimized compounds occupied a greater volume within the S4 and S5 subpockets, enhancing binding stability and affinity. Structural modifications during optimization contributed to increased favorable interactions with critical residues (Figure 6). These findings highlight the potential of I3C-1 and I3C-34 as promising candidates in addressing multiple strains of coronaviruses and supporting the development of effective therapies against future viral outbreaks.

3.8. Comparison of AI-Predicted $\Delta\Delta G$ and FEP Calculated $\Delta\Delta G$. In this section, we provide a detailed comparison of the $\Delta\Delta G$ values derived from our AI predictions with those obtained through rigorous physics-based FEP calculations, underscoring the necessity for integrating both methodologies into the drug discovery process. The AI model, characterized by its rapid screening capabilities, efficiently narrows extensive chemical libraries by filtering out compounds that exhibit a low predicted activity. This allows researchers to focus on a more manageable subset of candidates for further evaluation. However, it is crucial to acknowledge that the accuracy of these AI predictions is heavily dependent on the quality and comprehensiveness of the training data, which can sometimes limit the reliability of the predictions. Conversely, FEP stands out as a robust and precise method for estimating binding affinities based on fundamental physical principles. It serves as a vital validation step for the candidates selected by AI, providing a deeper and more accurate assessment of their potential binding interactions. Our comparative analysis demonstrated a PCC in the range of 0.5 to 0.54 between the AI-predicted $\Delta\Delta G$ and the FEP-calculated $\Delta\Delta G$ values (see Figure S12). While this indicates a moderate but meaningful relationship, it emphasizes the importance of employing both approaches in tandem to fully capture the nuances of binding interactions. Looking forward, the potential to incorporate FEP-calculated data into AI training represents a compelling opportunity for enhancing predictive accuracy. By integrating high-quality FEP results into the AI model, we can refine its ability to identify complex patterns and relationships, ultimately improving the selection of effective antiviral candidates. This iterative strategy, while not explored in our current study, could significantly advance drug discovery efforts by providing a more nuanced understanding of molecular interactions.

4. CONCLUSION

In conclusion, the past three coronavirus outbreaks (SARS in 2003, MERS in 2012, and COVID-19 in 2019) have posed significant threats to human health. To respond rapidly to possible future coronavirus outbreaks, the development of antiviral drugs is imperative. The main protease is a critical target for the development of antivirals. In this study, we first constructed a drug design framework that integrates a physics-based FEP and data-driven deep learning. This strategy was subsequently applied to optimize and modify quinoline-based inhibitors, resulting in the discovery of novel compounds with significantly enhanced binding affinities, such as I3C-1, I3C-2, I3C-34, I3C-35, and I3C-36 (see Figures 3 and 5 for compound structures). Overall, this study combines the advantages of physics-based MD simulations and data-driven deep learning, providing a virtual pipeline for identifying candidate drugs and identifying technical reserves for potential emergent coronaviruses.

■ ASSOCIATED CONTENT

Data Availability Statement

The source code of Deep-CovBoost can be accessed at https://github.com/yqyang733/Deep-CovBoost.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jcim.5c01307.

MM/GBSA energy decomposition (Table S1); Deep learning model performance (Tables S2 and S3); Predicted $\Delta\Delta G$ scores for analog design (Tables S4–S6); Comparison of FEP and experimental data (Figure S1); Molecular scaffolds (Figure S2); Deep learning screening results (Figures S3, S7, and S9); FEP results and MD-based interaction analyses (Figures S4, S6, S8, and S10); Structural comparisons with approved or reported inhibitors (Figures S5 and S11); AI-FEP correlation (Figure S12) (PDF)

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Notes

The authors declare no competing financial interest.

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